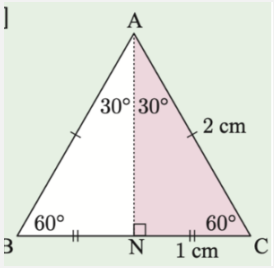
I- Radian

Investigation

Consider an equilateral triangle with sides 2 cm long. Altitude $[AN]$ bisects side $[BC]$ and the vertical angle $\widehat{BAC}$.

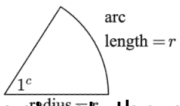


1. Can you use this figure to explain why $\sin 30^\circ = \frac{1}{2}$?
2. Use your calculator to find the values of $\sin 30^\circ$, $\sin 150^\circ$, $\sin 330^\circ$, $\sin 210^\circ$, and $\sin(-330^\circ)$.
3. Can you explain your answers even though the angles are not acute?

We have previously noted that a full revolution makes an angle of 360°, and the angle on a straight line is 180°. Hence, one degree, 1°, can be defined as $\frac{1}{360}$ of one full revolution. This measure of angle is often used by surveyors and seamen.

Definition

An angle is said to have a measure of 1 *radian* (1^c) if it is subtended at the center of a circle by an arc equal in length to the radius.



Remark. The symbol "rad" is used for radian measure but is usually omitted. By contrast, the degree symbol is always used when the measure of an angle is given in degrees.

From the diagram we can see that 1^c is slightly smaller than 60°; in fact,

$$1^c \approx 57.3^\circ.$$

The word *radian* is an abbreviation of "radial angle".

Property

$$\pi \text{ radians} = 180^\circ.$$

Hence,

$$\text{Degrees} \xrightarrow{\times \frac{\pi}{180}} \text{Radians} \quad \text{and} \quad \text{Radians} \xrightarrow{\times \frac{180}{\pi}} \text{Degrees}.$$

Example

1. Convert 45° to radians in terms of π.
2. Convert 126.5° to radians.
3. Convert $\frac{5\pi}{6}$ to degrees.

Exercise 1.

1. Convert to radians, in terms of π :

$$90^\circ,\ 135^\circ,\ 315^\circ,\ 60^\circ,\ 225^\circ,\ 30^\circ,\ 18^\circ,\ 360^\circ,\ 80^\circ.$$

2. Convert to radians, correct to 3 significant figures :

$$36.7^\circ,\quad 137.2^\circ,\quad 317.9^\circ,\quad 219.6^\circ.$$

Exercise 2.

1. Convert the following radians to degrees. Give your answers correct to 2 decimal places.
   (a) 2 (b) 1.53 (c) 0.867 (d) 3.179 (e) 5.267.
2. Copy and complete, giving answers in terms of π :

Degrees	0	45	90	135	180	225	270	315	360
Radians									

Degrees	0	30	60	90	120	150	180	210	240	270	300	330	360
Radians													

Definition

In the diagram, the arc length AB is l. Angle θ is measured in radians. We use a ratio to obtain :

$$\frac{\text{arc length}}{\text{circumference}} = \frac{\theta}{2\pi} \quad \Rightarrow \quad \frac{l}{2\pi r} = \frac{\theta}{2\pi} \quad \Rightarrow \quad l = \theta r.$$

For θ in radians, $l = \theta r$. For θ in degrees, $l = \frac{\theta}{360} \times 2\pi r$.

Definition

In the diagram, the area of minor sector XOY is shaded. θ is measured in radians. We use a ratio to obtain :

$$\frac{\text{area of sector}}{\text{area of circle}} = \frac{\theta}{2\pi} \quad \Rightarrow \quad \frac{A}{\pi r^2} = \frac{\theta}{2\pi} \quad \Rightarrow \quad A = \frac{1}{2}\theta r^2.$$

For θ in radians, $A = \frac{1}{2}\theta r^2$. For θ in degrees, $A = \frac{\theta}{360} \times \pi r^2$.

Example

A sector has radius 12 cm and angle 3 radians. Find its arc length and area.

Definition

Consider a circle with centre $(0,0)$ and radius r units. Suppose $P(x,y)$ is any point on this circle. Since $OP = r$,

$$\sqrt{(x-0)^2+(y-0)^2} = r \qquad \text{(distance formula)} \quad \Rightarrow \quad x^2+y^2 = r^2.$$

Thus $x^2+y^2 = r^2$ is the equation of a circle with centre $(0,0)$ and radius r. The equation of the *unit circle* is $x^2+y^2 = 1$.

Definition

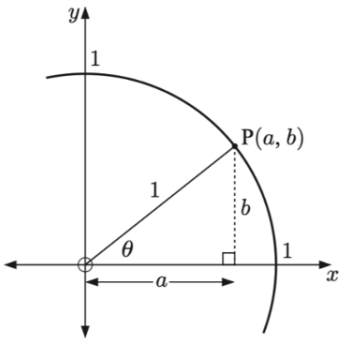
Suppose P lies anywhere on the unit circle and A is $(1,0)$. Let θ be the angle measured from $[OA]$ on the positive x-axis.

θ is *positive* for anticlockwise rotations and *negative* for clockwise rotations.

Example

$\theta = 210^\circ$ and $\phi = -150^\circ$.

II- Trigonometric ratios



Definition

$\cos\theta$ is the x-coordinate of P,

$\sin\theta$ is the y-coordinate of P.

Proof

Consider a point $P(a,b)$ which lies on the unit circle in the first quadrant. $[OP]$ makes an angle θ with the x-axis as shown.

Using right angled triangle trigonometry :

$$\cos\theta = \frac{\text{ADJ}}{\text{HYP}} = \frac{a}{1} = a, \qquad \sin\theta = \frac{\text{OPP}}{\text{HYP}} = \frac{b}{1} = b, \qquad \tan\theta = \frac{\text{OPP}}{\text{ADJ}} = \frac{b}{a} = \frac{\sin\theta}{\cos\theta}.$$

Example

Use a unit circle diagram to find the values of

$$\cos(-270^\circ) \quad \text{and} \quad \sin(-270^\circ).$$

Property

$$\cos^2\theta + \sin^2\theta = 1.$$

Proof

Since the unit circle has equation

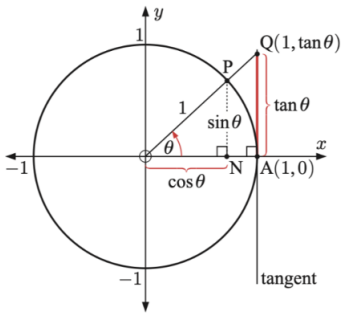
$$x^2 + y^2 = 1,$$

we obtain

$$(\cos\theta)^2 + (\sin\theta)^2 = 1 \quad \text{for all } \theta.$$

Definition

$$\tan\theta = \frac{\sin\theta}{\cos\theta}.$$



Proof

For all points on the unit circle,

$$-1 \leqslant x \leqslant 1 \quad \text{and} \quad -1 \leqslant y \leqslant 1.$$

So,

$$-1 \leqslant \cos\theta \leqslant 1 \quad \text{and} \quad -1 \leqslant \sin\theta \leqslant 1 \quad \text{for all } \theta.$$

Also, since

$$\frac{AQ}{OA} = \frac{NP}{ON}, \qquad \frac{AQ}{1} = \frac{\sin\theta}{\cos\theta},$$

Remark.

Quadrant	$\cos\theta$	$\sin\theta$	$\tan\theta$
1	positive	.....	.....
2	.....	.....	.....
3	.....	.....	.....
4	.....	.....	.....

Example

Find exactly the possible values of $\cos\theta$ for $\sin\theta = \frac{2}{3}$. *Illustrate your answers.*

Example

If $\sin\theta = -\frac{3}{4}$ and $\pi < \theta < \frac{3\pi}{2}$, find $\cos\theta$ and $\tan\theta$. *Give exact values.*

III- Applications

Property

For θ in radians :

$$\sin(\pi - \theta) = \sin\theta$$

$$\cos(\pi - \theta) = -\cos\theta$$

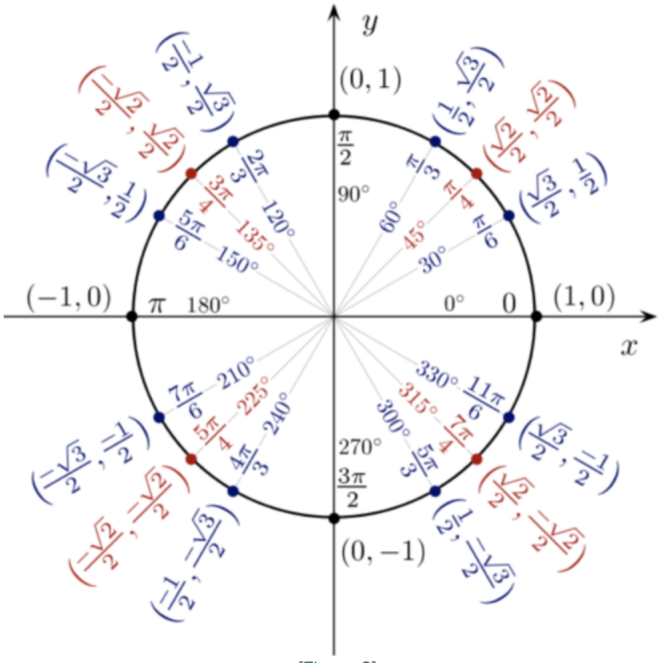
$$\cos(2\pi - \theta) = \cos\theta$$

Example

Find the two angles θ on the unit circle, with $0 \leqslant \theta \leqslant 2\pi$, such that :

(a) $\cos\theta = \frac{1}{3}$, (b) $\sin\theta = \frac{3}{4}$, (c) $\tan\theta = 2$.

Property



Property

For $\theta \in \mathbb{R}$, $k \in \mathbb{Z}$

$$\cos(\theta + 2k\pi) = \cos\theta$$

$$\sin(\theta + 2k\pi) = \sin\theta$$

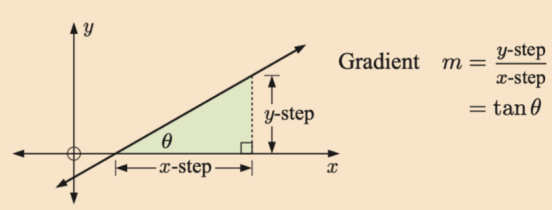
$$\tan(\theta + k\pi) = \tan\theta$$

Property

If a straight line makes an angle of θ with the positive x-axis, then its gradient is

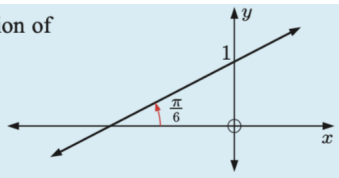
$$m = \tan\theta$$

Proof



Example

Find the equation of the given line:



IV- Differentiation

Suppose P moves anticlockwise around the unit circle with constant linear speed of 1 unit per second.

After 2π seconds, P will travel 2π units which is one full revolution, and thus through 2π radians.

So, after t seconds P will travel through t radians, and at time t, P is at $(\cos t, \sin t)$.

Definition

The angular velocity of P is the rate of change of $\angle A\hat{O}P$ with respect to time.

Remark. Angular velocity is only meaningful in motion along a circular or elliptical arc.

The angular velocity of P is $\dfrac{d\theta}{dt}$ and

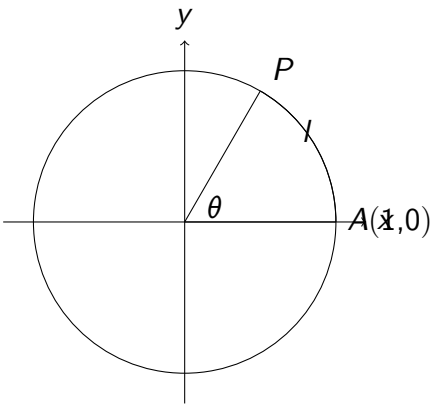
$$\frac{d\theta}{dt} = 1 \text{ radian per second.}$$

Property

If we let l be the arc length AP, the **linear speed** of P is $\dfrac{dl}{dt}$, the rate of change of l with respect to time.

Proof

$$l = \theta r = \theta \times 1 = \theta \quad \text{and} \quad \frac{dl}{dt} = 1 \text{ unit per second.}$$



Property

For x in radians :

If

$$f(x) = \sin x \text{ then } f'(x) = \cos x$$

If

$$f(x) = \cos x \text{ then } f'(x) = -\sin x$$

Property

$$\text{If } f(x) = \tan x \text{ then } f'(x) = \frac{1}{\cos^2 x}$$

Proof

$$y = \tan x = \frac{\sin x}{\cos x}$$

We let $u = \sin x$ and $v = \cos x$.

$$\therefore \frac{du}{dx} = \cos x \quad \text{and} \quad \frac{dv}{dx} = -\sin x$$

$$\therefore \frac{dy}{dx} = \frac{u'v - uv'}{v^2} \quad \{\text{quotient rule}\}$$

$$= \frac{\cos x \cos x - \sin x(-\sin x)}{(\cos x)^2}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x} \quad \{\text{since } \sin^2 x + \cos^2 x = 1\}$$

Remark.

$$\frac{1}{\cos x} \text{ is often called secant } x \text{ or } \sec x, \text{ so } \frac{d}{dx}(\tan x) = \sec^2 x.$$

Property

Function	Derivative
$\sin[f(x)]$	$\cos[f(x)]\, f'(x)$
$\cos[f(x)]$	$-\sin[f(x)]\, f'(x)$
$\tan[f(x)]$	$\dfrac{f'(x)}{\cos^2[f(x)]}$

Proof

Suppose $y = \sin[f(x)]$

If we let $u = f(x)$, then $y = \sin u$.

But

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \{\text{chain rule}\}$$

$$\therefore \frac{dy}{dx} = \cos u \times f'(x)$$

$$= \cos[f(x)] \times f'(x)$$

We can perform the same procedure for $\cos[f(x)]$ and $\tan[f(x)]$.

Example

Differentiate with respect to x :

1. $x \sin x$
2. $4\tan^2(3x)$